CERN Bulletin

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http://home.cern/cern-community

A year to celebrate CERN's 70th anniversary

Celebrations begin with Unveiling the Universe, an event combining science, art and culture in CERN Science Gateway on 30 January

Happy 2024 from CERN!

In 2024, CERN is celebrating its 70th anniversary. Throughout the year, events and activities will showcase the Laboratory's rich past, as well as its bright future with a unique accelerator complex set to drive 70 more years of research into what the Universe is made of and how it works.

With the full programme of events soon to be announced, join us for the first event, Unveiling the Universe, on **30 January** in [CERN Science Gateway](). This event will combine science, art and culture, looking back on CERN's 70 years of scientific discoveries and looking forward to how many mysteries of the Universe are still to be understood.

Physicists and artists will explore human creativity, curiosity, imagination and inspiration, looking at how we have built and will continue to build knowledge and understanding of the big questions about the Universe: from the first billionth of a second after the Big Bang to the invisible "dark" Universe.

The day will begin with the inaugural CERN Art and Science Summit. Renowned artists and scientists will delve into the challenges and opportunities of engagement between art and science. The summit will highlight the achievements of CERN's forward-thinking approach to arts and creativity through the [Arts at CERN]() programme.

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Happy 2024 to the CERN community

2024 will be a very special year for CERN, as the Lab will turn 70.

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A Word from Fabiola Gianotti

Happy 2024 to the CERN community

Happy New Year to you all, I wish you and your families and friends a wonderful 2024.

2024 will be a very special year for CERN, as the Lab will turn 70. This is a remarkable milestone, and we will be celebrating it with our partner institutes and all our collaborators in our Member States, Associate Member States and beyond.

Throughout the year, we will demonstrate how, over the past seven decades, CERN has been at the forefront of scientific knowledge and technological innovation, and a model for training and education, collaboration, open science and inspiration of citizens around the world.

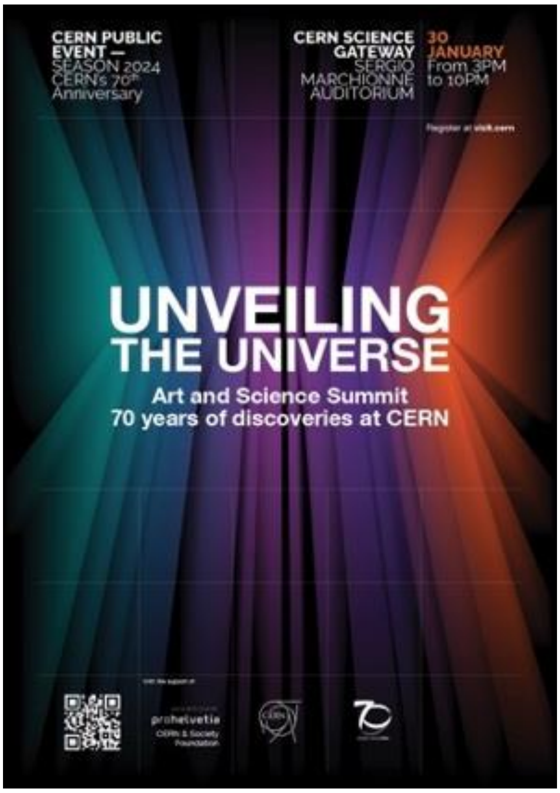
We are the proud stewards of an extraordinary heritage, which we will honour in many ways. But this anniversary is also a great opportunity to look forward. We will be stoking the enthusiasm of people of all ages by showing how CERN's beautiful journey of exploration into the fundamental laws of nature and the fundamental constituents of matter and the Universe is set to continue into the future.

This year's celebrations are also very much a recognition of CERN's strongest asset, its personnel and its worldwide community – this means each and every one of you, and the thousands of people who, with their work through the decades, have made CERN a unique place. Today, we look back at CERN's great accomplishments with pride and look forward to its bright future with enthusiasm.

Fabiola Gianotti
CERN Director-General

A video message from Fabiola Gianotti to the CERN community is available here: https://videos.cern.ch/record/2299470.

The CERN Directorate's New Year presentation to personnel will take place on Thursday 25 January at 10 a.m. in the Main Auditorium. More information on Indico: https://indico.cern.ch/event/1367641/.



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The evening will be one of conversations with distinguished scientists of different generations – David Gross, Djuna Croon, Gian Francesco Giudice and Tara Shears – exploring the development of knowledge in particle physics and CERN's contributions to these developments.

All this will be interspersed with a visit to the art-based Exploring the Unknown exhibition in CERN Science Gateway, and the evening will be rounded off with Enigma, a visual and audio performance.

For more information and to register, visit: https://indico.cern.ch/event/1360288.

CERN70: Foundations for European science

Part 1 of the CERN70 feature series : François de Rose, a French diplomat, was involved in the creation of CERN. In 2004, he still remembered the first discussions that ultimately led to the birth of the Organization

Many of CERN's founders gathered for the third session of the provisional CERN Council in Amsterdam on 4 October 1952. At this session, Geneva was chosen as the site for the Laboratory and it was decided to build a 25-30 GeV Proton Synchrotron. (Image: CERN)

In the aftermath of the Second World War, a handful of visionary scientists imagined how to revive science in Europe. By pooling the resources of several countries, they hoped to equip Europe with accelerators similar to those being built in the United States, and thus stem the brain drain. The idea of creating a European atomic physics laboratory took shape. After months of negotiations, the UNESCO Intergovernmental Conference in 1951 adopted the first resolution to create a European Council for Nuclear Research (CERN).

The CERN Convention, drawn up in 1953, was gradually ratified by the 12 founding Member States: Belgium, Denmark, France, the Federal Republic of Germany, Greece, Italy, the Netherlands, Norway, Sweden, Switzerland, the United Kingdom and Yugoslavia.

On 29 September 1954, the European Organization for Nuclear Research officially came into being. The provisional Council was dissolved, but the acronym remained.

Recollections

François de Rose, a French diplomat, was involved in the creation of CERN from the very beginning. He went on to hold office as President of the CERN Council from 1958 to 1960, during which time he helped to prepare the Laboratory's extension into

French territory. Interviewed in 2004, he had vivid memories of the initial discussions that led to the birth of the Organization.

François de Rose (left) with CERN Director-General John Adams at the inauguration of the PS in 1960. (Image: CERN)

"CERN is one of the achievements with which I am the most proud to have been associated … it is such a noble cause.

The first steps towards CERN's creation were taken in the United States between 1947 and 1949. At that time I was the French representative to the United Nations International Atomic Energy Commission, which comprised both diplomats and scientists. It was there that I met Robert Oppenheimer, with whom I struck up a friendship. Like many American scientists, he had been very much influenced by European science, having worked in Niels Bohr's group in particular. During one of our conversations he said more or less the following: "We have learnt all we know in Europe. But in the future, fundamental physics research is going to require substantial resources which will be beyond the means of individual European countries. You will need to pool your efforts to build these big machines that are going to be needed. It would be unhealthy if the Europeans were obliged to go the United States or the Soviet Union to conduct their fundamental research." The idea fascinated me and I arranged for him to meet the French scientific advisers from my Commission, Pierre Auger, Francis Perrin, Lew Kowarski, and Bertrand Goldschmidt.

In 1949, when we returned to Paris, I went on a tour of European capitals with Francis Perrin to see what sort of reception Oppenheimer's idea would be given. We were confronted with a lack of interest: the scientists were afraid that a big research centre would swallow all the available funds and soak up the resources of their own laboratories. They were wrong, however, because as soon as CERN started to request resources there was an increase in the funding allocated to research. What's more, the governments had no idea of what it was all about: when they heard the words 'atomic research', they immediately thought of the atom bomb and were afraid that it would not go down well with the Americans. Last but not least, the fact that Frédéric Joliot Curie, an eminent member of the Communist Party, was in charge of the French Atomic Energy Commission caused the other European scientists to have cold feet. We therefore failed in our mission. However, the idea was now on the table and Isidor Rabi's speech at the Florence General Conference secured the breakthrough we needed.

CERN was created so that Europeans were not forced to go the United States. Today, Americans are coming to Europe to work on CERN's machines, something which I don't think Oppenheimer had anticipated. I find that an extraordinary turnaround."

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. François de Rose passed away in 2014 at the age of 103, read more about him in the CERN Courier.

CERN highlights in 2023

From high above to far below, journey through the CERN highlights of 2023

In 2023, CERN celebrated a year of achievements on its journey of scientific exploration. The inauguration of CERN Science Gateway, an emblematic education and outreach centre, reflected CERN's commitment to inspiring future generations.

Precision measurements took centre stage as the ATLAS experiment set records in studying the Higgs boson mass and the strong force strength. The CMS experiment presented its search for dark photons and other exotic particles and measured tau-lepton polarisation in Z-boson decays. The ALICE experiment shone light on the nucleus by probing its intricate structure and on charm and beauty quark dynamics in quark–gluon plasma. The LHCb experiment made the most precise measurement yet of matter–antimatter asymmetry with beauty quarks and observed hypertriton, a key to modelling neutron star cores. Lead ions collided in the Large Hadron Collider for the first time in five years and collider neutrinos were observed for the first time by FASER and SND@LHC. To prepare for the future, the High-Luminosity LHC team completed and validated key hardware components, underground structures and services. From observing antimatter's fall under gravity to advancements in atomic clock precision and probing dark matter searches, CERN's diverse experiments underlined the Laboratory's key role in shaping the future of particle physics and our understanding of the Universe.

CERN celebrated the Year of Open Science with NASA and other scientific institutions and collaborated with the International Committee of the Red Cross and the World Food Programme to use its technologies for humanitarian action and fighting hunger. CERN's impact reached far beyond the Laboratory with initiatives such as exploring superconductivity solutions for renewable energy transmission with SuperNode, moving forward with the demonstrator for low-carbon aviation with Airbus and accelerating deep-tech startups with CERN Venture Connect.

Host State presidents Emmanuel Macron and Alain Berset also visited CERN this year.

Watch CERN in 2023 and enjoy a visual journey through some of the many highlights of the year, immersing yourself in groundbreaking achievements and scientific wonders.

Watch the video here:
https://videos.cern.ch/record/2299435.

CERN Council decides to conclude cooperation with Russia and Belarus in 2024

The Organization's cooperation with the Russian Federation and the Republic of Belarus will conclude at the expiry in 2024 of the International Cooperation Agreements (ICAs)

Following a decision taken by the CERN Council on 15 December, the Organization's cooperation with the Russian Federation and the Republic of Belarus will conclude at the expiry in 2024 of the International Cooperation Agreements (ICAs) with the two countries. This decision was taken following the Council's June 2022 decisions.

International Cooperation Agreements with the Organization normally run for five years, and are tacitly renewed for the same period unless written notice is provided by one party to the other at least six months prior to expiration.

CERN Council Meeting in December 2023. (Image: CERN)

The cooperation will come to an end on 27 June 2024 for the Republic of Belarus and on 30 November 2024 for the Russian Federation. All relations between CERN and Russian and

Belarusian institutions will cease as of these dates. Relations continue with scientists of Russian or Belarusian nationality otherwise affiliated with CERN.

The Council reaffirmed that all the decisions it has taken to date in light of the ongoing military invasion of Ukraine by the Russian Federation with the involvement of the Republic of Belarus, along with the actions undertaken by the CERN Management, remain in force. These include measures adopted at the extraordinary meeting of the Council on 8 March 2022, and at the Council's regular meetings on 25 March 2022 and 16 June 2022.

The full text of the Council's resolutions can be found here.

Work Well Feel Well: A break that revitalises you?

Part 2 of the Work Well Feel Well series looks at why taking a break is so important

Time to take a break

As one year ends and another begins, the two-week CERN closure can give us time to rest. **Taking a break is necessary**, not only at the end of the year, but **within each working day**. We can all make it a New Year's resolution in 2024 to find time to take a break.

Why? Allowing ourselves to rest aids our **physical and mental wellbeing**. We return to a task with **renewed energy and perspective**, helping to **increase our efficiency** in the short and medium term.

Our workdays have an **official break of one hour for lunch**. Let's **reclaim it**. It is an official time where we can **disconnect from work** and **reconnect with colleagues**. The pause **restores vitality** and improves our wellbeing. We can also add **micro-breaks** into our days, helping to keep a healthy distance from ongoing tasks; for example taking our eyes away from the computer screen for a few minutes.

Some breaks are almost invisible: a long deep breath before continuing to work. Other breaks involve a pause in concentration, setting aside a complex task to work on a simpler one. **A break doesn't always mean doing nothing**; it can also involve doing something else. In fact, some of us need to do something to clear our minds.

CERN70: Some of CERN's first personnel take a break in 1958, with the Salève mountain in the background. (Image: CERN)

A break that revitalises you?

We've all taken breaks that didn't help to recover our strength. Even worse, we may return to work more tired than before. The best way to see if the break has been beneficial is to see how we feel when we come back.

A real break replenishes our energy and **increases our efficiency**. This means **quietening our minds**, changing our ideas and stepping back from current activities. It's all about **responding to our own needs**: recharging our batteries, exercising, eating and drinking properly. It's also knowing how to recreate internal availability, often lost among

unfinished tasks and the demanding rhythm of work.
Offering ourselves a beneficial break, be it official or regular micro-breaks, involves **working out what we really need and allowing ourselves that time**.

Take action

As part of the "Efficiency and caring at work" campaign, the Work Well Feel Well website now offers useful resources that can be downloaded, including self-reflection exercises and sleep advice.

- The recordings of the recent talks by the CERN Medical Service and psychologists on the topics of "Flash disconnect" and "Cardiac coherence" provide practical advice and exercises to take a break during the working day.
- The recording of the 2019 CERN talk "Sleep: The Wake Up Call" by Vicki Culpin highlights the importance of resting.
- For those who prefer a break to be a chance to do something else, the Staff Association provides a list of wide-ranging clubs.

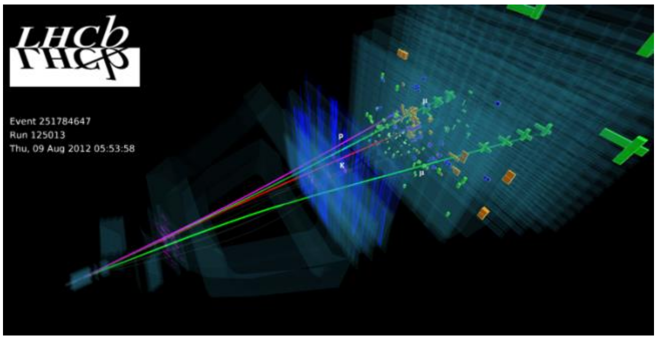
This is the second of a 12-part Work Well Feel Well series, with articles to be published every two months.

HR department

LHCb experiment releases all of its Run 1 proton–proton data

The latest release makes LHCb research data, used by researchers to produce a number of significant results, available to anyone for a wide range of physics studies

At the end of December 2023, the LHCb experiment released all its data from Run 1 of the Large Hadron Collider. This data, collected by the experiment in 2011 and 2012, contains approximately 800 terabytes of information obtained from proton–proton collisions. The data has been made available in a pre-filtered format, suitable for a wide range of physics studies for research and education purposes.
LHCb data across Runs 1 and 2 has already been used for over 700 scientific publications, including numerous significant findings. All results from the collaboration have already been made publicly accessible in open-access papers and the numerical results from the graphs can be consulted in the HEPData database. With the new release, the data used by the researchers to produce these results is now accessible. The data has been released in the framework of CERN's Open Data Policy, which reflects values that have been enshrined in the CERN Convention for more than sixty years and applies to all of CERN's activities.



An event display recorded in 2012, during LHC Run 1. (Image: LHCb collaboration)

The collaboration has pre-processed the data by reconstructing experimental signatures, such as the trajectories of charged particles, from the raw information delivered by the complex detector system. The data is filtered, classified according to a large number of processes and decays, and made available in the same format that is used internally by LHCb physicists. The data can be downloaded from the CERN Open Data portal.
To aid the user's understanding, the samples come with extensive documentation and metadata, as well as a glossary explaining several hundred specialised terms used in the pre-processing. The data can be analysed using dedicated LHCb

algorithms, which are available as open-source software.
All data sets have digital identifiers (DOIs) for reference and citation. The experiment also welcomes feedback on how the data is used and invites users to discuss and post questions in the CERN Open Data Forum.

LHCb collaboration

Hi-Lumi News: First magnet from the US Accelerator Upgrade Project shipped to CERN

Watch a timelapse of the unboxing of the first of ten cryo-assemblies from the US to be installed in the HL-LHC interaction regions

The cryo-assembly containing two MQXFA magnets was delivered to CERN in early December. (Image : CERN)

At the beginning of December, CERN received an important shipment. It contained a cryo-assembly of two 4.2-m-long magnets developed by the Accelerator Upgrade Project in the US. These magnets are vital for the high-luminosity upgrade of the LHC (HL-LHC). With coils made from niobium–tin, instead of the niobium–titanium that the LHC currently uses, they will help focus the particle beams to an even smaller spot size at the interaction points of the ATLAS and CMS experiments.
This is the first of ten cryo-assemblies that will make the month-long journey from the US. A celebration was held at CERN on Monday, 18 December to commemorate this milestone, bringing people from both sides of the Atlantic together. "In the realm of large scientific endeavours like the HL-LHC, global collaboration and expertise play pivotal roles. The delivery of the first cryo-assembly housing fully validated niobium–tin series magnets is a tangible testament to the success of the US Accelerator Upgrade Project," says Mike Lamont, CERN Director for Accelerators and Technology. "This event not only marks a crucial milestone in our collaboration with our US partners, but also celebrates the outstanding contributions shaping the future landscape of particle physics at CERN."
These cryo-assemblies from the US will be used in conjunction with the 7.2-m-long magnets of the same design developed at CERN for the HL-LHC. Both types of magnet are to be installed in the inner-triplet (IT) string, a facility at CERN built to test all the components that will comprise the interaction regions at ATLAS and CMS.
"With this arrival, five out of the six cryomagnets for the IT string are now at CERN ready for testing," says Ezio Todesco, who is in charge of the HL-LHC interaction region magnets. "The string will take on good colours in 2024."
"This first cold-mass assembly will go into the IT string and will be extensively tested throughout 2025," says Oliver Brüning, HL-LHC project leader. "This marks the start of a new phase of the CERN-US collaboration: the delivery of final cryo-assemblies ready for installation in the LHC."
Giorgio Apollinari, Project Director of the HL-LHC AUP, is looking forward to this next phase of the project. "We have been building these magnets in the US for the last five years and have now completed 75% of their production," he said. "We've sent the first two magnets in a final cryo-assembly after extensive tests in the US and are already putting together the next cryo-assembly, scheduled to arrive at CERN by early 2024."
Watch a timelapse of the magnet unboxing at CERN below.

Naomi Dinmore

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Watch the video here: [https://youtu.be/st-wb7GNH4c.](https://youtu.be/st-wb7GNH4c)

Shaping future quantum techniques in machine learning at CERN

The latest advancements in quantum techniques in machine learning have been presented at CERN, showcasing applications in particle physics and beyond

Natalia Ares from the University of Oxford presents machine learning for tackling quantum device variability at the 7th International Conference on Quantum Techniques in Machine Learning, held at CERN in November. More than 300 people attended in person, with more following online. (Image : CERN)

Problem solving gets faster if quantum methodologies are used instead of classical computers. Physicists and computer scientists are therefore working closely together to explore this potential. In November, the 7th International Conference on Quantum Techniques in Machine Learning (QTML) was held at CERN, bringing together more than 300 researchers and industry partners in the field.

Machine learning uses data and algorithms to help computers to learn patterns and perform more effectively at tasks ranging from helping doctors to diagnose cancer to improving facial recognition. Combining techniques from quantum physics with machine learning can reduce the number of steps needed for algorithms to obtain a correct answer.

"CERN is putting significant effort into developing quantum technology for particle physics and beyond, through the Quantum Technology Initiative and the Open Quantum Institute," explained Alberto di Meglio, head of the Innovation section in CERN's IT department, in his opening speech. As well as talks from researchers, organisations and companies showcased their latest developments, with talks from ESA, Google, IBM, Intel, IONQ, NASA and PASQAL. Applications ranged from optimising aircraft cargo loading to developing new algorithms to study lithium compounds and their chemical reactions in battery chemistry. "The presence of major industry partners was a key element of the conference," confirms Michele Grossi, senior fellow in quantum computing and algorithms at CERN. "The continuous interaction between industry and academia is helping the community to drive the quantum revolution in a fair way."

The conference was organised without parallel sessions, which divide participants, enabling researchers from various fields to interact. "This conference allowed more than 300 people to gather each day to exchange around one focused theme," says Miguel Marquina, senior staff member in CERN's IT department. "It is powerful to experience such an engaging environment."

The 8th edition of the International Conference on Quantum Techniques in Machine Learning will take place in 2024 at the University of Melbourne. More information can be found on the QTML 2023 and QTI websites.

Mariana Velho

Serbia joins the Worldwide LHC Computing Grid

On 9 December, CERN and Serbia signed a Memorandum of Understanding (MoU) at the Serbian State Data Centre. The centre will become a Tier1 member of the Worldwide LHC Computing Grid (WLCG), the highest level of collaboration within the Grid

The WLCG is a network of computing centres distributed across more than 40 countries that provides global computing resources for the storage, distribution and analysis of the data generated by the LHC.

This complex network is organised in tiers: Tier 0 is the CERN Data Centre; Tier 1 centres are powerful computing farms capable of providing permanent storage of raw data; and Tier 2 centres are additional data centres located in various collaborating institutes worldwide. The MoU signed on 9 December in Serbia marks a decisive step towards the country becoming a WLCG Tier 1 centre. The Serbian data centre will initially receive data from the CMS experiment, of which Serbia is a member.

"The signature of this MoU is a testimony to the vision of the Serbian government and the Serbian scientific community and their commitment towards the CERN research programme, and vice versa," said Enrica Porcari, head of the CERN IT department, who signed the MoU. "We are impressed by the computing infrastructure that Serbia is making available to the Worldwide LHC Computing Grid, and, in particular, to the CMS collaboration. I am sure that this MoU will consolidate even further the longstanding relationship between CERN and Serbia and will contribute to realising our common vision to become a springboard for scientific collaboration here and worldwide. CERN continues to be committed to the co-development and sharing of knowledge with its MoU partners in the spirit of scientific progress."

From left to right: Enrica Porcari, Head of CERN's IT department, Jelena Begović, Serbian Minister of Science, Technological Development and Innovation, and Mihailo Jovanović, Serbian Minister of Information and Telecommunications. (Image: Serbian Ministry of Information and Telecommunications)

Antonella Del Rosso

CERN Alumni Third Collisions: A confluence of reconnection, networking and celebration

Join us at the Third Collisions event, "Accelerating Beyond", taking place at CERN on 9 – 11 February

"Great stories of entrepreneurship! It was amazing to see the impact that CERN alumni are making in society." These are the words of a CERN alumni who took part in the Second Collisions event, in 2021. These sentiments echo the anticipation surrounding the upcoming Third Collisions event, taking place at CERN from 9 to 11 February.

In less than a month, CERN will play host to several hundred alumni, marking the third reunion of this network, now entering its seventh year of existence. The event promises a rich tapestry of experiences, blending insightful talks, stimulating panels, and the chance to reconnect with former colleagues. The focus of the talks and panels extends beyond the scientific realm, delving into pressing topics like climate change, artificial intelligence, and quantum computing. The event serves as a nexus for networking, providing opportunities for alumni to reconnect and engage with members of the CERN community.

A notable addition to this year's programme is the inaugural jobs fair, bringing together companies and EIROforum organisations actively seeking individuals with skills cultivated at CERN. CERN Alumni companies will also feature, as will CERN teams such as Knowledge Transfer, CERN Courier, and the CERN & Society Foundation, amongst others. Many companies will be in place, as of the afternoon of Thursday, 8 February, in the Main Building, and will eagerly welcome those at CERN contemplating their next professional venture.
The Third Collisions aren't just about work; they encompass a spectrum of events, sparking CERN nostalgia, including a welcome reception, gala dinner, and multiple coffee breaks that facilitate informal networking. The agenda also boasts diverse entertainment options, ranging from a ski outing to a MusiClub alumni DJ set and the screening of "Almost Nothing", followed by a Q&A with one of the directors. Fitness activities are also on the menu, ensuring a well-rounded experience for all participants.
The event's detailed programme, list of speakers, and additional information can be accessed here.
To facilitate participation, a small financial contribution is requested, ensuring the smooth execution of this grand event. For the first time, participants are welcome to bring a plus one (over 18s only), enhancing the sense of community and celebration.
Additionally, the organising committee is seeking volunteers to contribute to the event's success. Those willing to lend a helping hand will have their registration fees waived, further emphasising the collaborative spirit that defines the CERN and CERN Alumni communities. Please contact alumni.relations@cern.ch for more information.
Join us at CERN Alumni Third Collisions, which promise to be an extraordinary blend of knowledge exchange, professional networking, and celebration, reinforcing the impact of CERN alumni on society and echoing the continuous legacy of excellence set by CERN and its remarkable alumni.

CERN Alumni programme

Computer Security: Hits are coming closer

Like it or not, the cyber realm is unfortunately developing alongside the physical world. While in the real world, conflicts tragically dominate world politics, the usual commercial cyber-attackers have increased their attacks, too. And, unlike in the past, the research and education (R&E) sector is no longer spared.
Until recently, universities had been attacked only very occasionally. One of the last in this sad line-up was the University of Michigan. But the last year has also seen major attacks against accelerators and telescopes – as collateral damage and as the attackers' main focus, respectively. What we feared in the past has become reality: where there is operational value, there is a business opportunity for malicious evildoers to extract money – and this applies to the R&E community as well. To "ransom" the operator, threaten operations, stop production and cause damage.
Over the past 12 months, the CERN Computer Security team has tirelessly helped dozens of universities worldwide to protect themselves against such "ransomware" attacks (see our monthly security reports) and improve their defences, as well as providing training, tipping

them off to imminent danger where our threat intel permits and assisting them in incident response when it was too late and damage had been done. Similarly, the base question is not "if" but "when" CERN will be subject to a ransomware attack. The three mantras of ransomware defence are "Don't get it", "Don't pay" and have all-encompassing, complete and thoroughly tested back-ups in place. While CERN has taken a firm position on the second mantra (incidentally, governments are increasingly prohibiting ransom payments), and the IT department, in collaboration with many stakeholders in the Organization, is hard at work on the third, the first mantra – raising our defences – is the hardest one. Many projects are already under way and we're not done yet:

- 2024 will see an even more all-encompassing roll-out of 2-factor protection, in particular to our user community and to holders of so-called "secondary" accounts. It will eventually also cover LXPLUS and the CERN Windows terminal servers.
- The "new" anti-malware solution will finally be deployed to all CERN/centrally managed Windows PCs and we will investigate whether this protective means can also be forced onto any other Windows laptop or Macbook purchased and owned by the Organization.
- Vulnerability scanning and penetration testing against CERN's internet presence is currently being tendered and will start in early 2024 (the owners of vulnerable websites and web servers may possibly be required to contribute to the cost).
- Together with HR's Learning and Development group, we will expand CERN's training catalogue and offer dedicated hands-on courses on secure programming and software development, as well as IT operations.
- In parallel, Gitlab security scanning may make it into your pipelines and into your choice of virtual machines and containers in order to reduce the risk through the "software supply chain".
- Our Security Operations Centre will extend its remit to cover even more data sources, enabling us to monitor more network segments than ever before, as well as our main cloud-based services (such as Google and Microsoft).
- Finally, CERN recently concluded an external audit on "cyber security" and its findings and resulting recommendations will be addressed in the course of 2024 (more on that in a future Bulletin article).

In any case, ransomware hits are coming closer. Unlike some of our unfortunate partner universities and some astronomy experiments and particle accelerators, CERN has not been hit yet. Yet! And we hope to keep it that way. Cybersecurity is a permanent marathon: our work will never be done. But for this race, we appreciate (and need!) your help in securing the Organization. As "sec_irty" is not complete without "u"!!! Let's have a (more) peaceful 2024.

CERN Computer Security team

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In case you missed it in December, the talk "CERN Computer Security: Abuse, Blunder and Fun" will be repeated on 30 January at 11 a.m. in the Council Chamber. More information on Indico: https://indico.cern.ch/event/1365440/.

Official news

CERN remuneration calendar in 2024

To all members of the personnel in receipt of remuneration from CERN.
In 2024, net monthly remuneration will be paid into individual bank accounts on the following dates:

- Thursday 25 January
- Monday 26 February
- Monday 25 March
- Thursday 25 April
- Friday 24 May
- Tuesday 25 June
- Thursday 25 July
- Monday 26 August
- Wednesday 25 September
- Friday 25 October
- Monday 25 November
- Thursday 19 December

FAP department

Pensions payment dates in 2024

- Monday 8 January
- Wednesday 7 February
- Thursday 7 March
- Monday 8 April
- Tuesday 7 May
- Friday 7 June
- Monday 8 July
- Wednesday 7 August
- Friday 6 September
- Monday 7 October
- Thursday 7 November
- Friday 6 December

HR department

Annual adjustments to financial benefits with effect from 1 January 2024

In accordance with recommendations made by the Finance Committee and decisions taken by Council in December 2023, the following financial benefits have been adjusted, with effect from 1 January 2024:

- A 0.29% increase to the scale of basic salaries paid to Staff Members and the scale of stipends paid to Fellows and Graduates (Annexes R A 5 and R A 6 of the Staff Regulations).
- A 1.3% increase to the subsistence allowances (Annex R A 7 of the Staff Regulations), family, child and infant allowances (Annex R A 3 of the Staff Regulations) and to payment ceilings of education fees (Annex R A 4 of the Staff Regulations).

The index of 7.8% for home leave has already been applied as of June 2023.
The new ceiling of education fees will be applied to the entire school year 2023-2024.
Other related adjustments have been implemented wherever applicable to associated members of the personnel (2024 subsistence rates).

The amended text of the Staff Regulations will be available shortly here: CERN Staff Rules and Regulations.

HR department

CERN Health Insurance Scheme (CHIS) – Monthly contributions as of 1 January 2024

The CHIS contribution rates are unchanged for 2024, but the contributions themselves will evolve due to the change in the relevant Reference Salary (see Chapter XII of the CHIS Rules). Thus, as of 1 January 2024, the lump-sum monthly contributions based on Reference Salary II will be as follows:

1. **Lump-sum contributions for voluntary members**

The monthly contribution for voluntary members (e.g. users and associates) with the normal health insurance will be 1 272 CHF per month, whilst for those with the reduced health insurance it will be 636 CHF.

2. **Lump-sum contributions for post-compulsory members other than CERN pensioners**

For post-compulsory members other than CERN pensioners, the monthly contribution will be 1 358 CHF in the case of former staff members and former spouses continuing their affiliation, whilst in the case of formerly dependent children continuing theirs it will be 543 CHF.

HR department

Staff Rules and Regulations – 11th edition

In accordance with recommendations made by the Finance Committee and decisions taken by Council in December 2023 (CERN/FC/6747-CERN/3770), please find below the pages of the Staff Rules and Regulations which have been updated further to the modifications coming into force on 1 January 2024:

Chapter II – Conditions of employment and association

Section 4 – Leave (*modification of page 25*)

Chapter V – Financial conditions

Section 1 – Financial benefits (*modification of page 45*)

Annex R A 3 - (*modification of page 68*)

Annex R A 4 - (*modification of page 69*)

Annex R A 5 - (*modification of page 71*)

Annex R A 6 - (*modification of page 72*)

Annex R A 7 - (*modification of page 73*)

The complete updated electronic version of the Staff Rules and Regulations is accessible via CDS.

HR department

Duty travel: Egencia tool from 1 January 2024

CERN's new duty travel solution, Egencia, launched CERN-wide on 1 January 2024

The new online booking tool includes the usual services such as flight, car rental and train booking as well as a new solution for accommodation. Travellers will now be able to book and manage their hotel reservations online. Hotels booked in Egencia will be pre-paid by CERN. A welcome e-mail was sent to all travellers on 2 January. Travellers are encouraged to log in and complete their profiles whether they plan on booking themselves or through their travel arrangers.

Training sessions are available for all members of the personnel; you can find them on Indico. Departmental travel managers can be contacted for more information on these sessions.
Members of the personnel shall use Egencia to book their future duty travels.

Questions can be addressed to the Duty Travel Coordination service.

FAP department

Reminder: Prohibition of certain laser pointers in Swiss territory

In Switzerland, the Federal Act on Protection against the Risks associated with Non-Ionising Radiation and with Sound (O-NIRSA), which came into effect on 1 June, 2019, prohibits laser pointers deemed dangerous. Considered as such lasers classified as 1M, 2, 2M, 3R, 3B and 4.
As a result, **these pointers may be confiscated at customs and destroyed**, even though they are not prohibited items in the cabin or hold of airplanes. Swiss authorities also remind that passengers in possession of such devices will be questioned by the police and may face criminal charges.
Only class 1 lasers, which pose no risk to health, are permitted.
More information:

- https://home.cern/news/official-news/cern/ban-laser-pointers-switzerland
- https://home.cern/news/announcement/cern/swiss-ban-laser-pointers-how-return-them-appropriate-disposal
- https://home.cern/news/announcement/cern/reminder-swiss-ban-laser-pointers

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Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

15–16 Jan 12:30 | At CERN | Building 664 | Jardin des Particules crèche et l'école | EN and FR
Le Jardin des Particules open days | Portes Ouvertes

16 Jan 09:00 | At CERN | Council Chamber and online | HR and Pension Fund | EN
Public presentation on CERN's pre-retirement programmes and pension-fund factors

19 Jan 09:00 | At CERN | Main Auditorium and online | HR and Pension Fund | FR
Présentation publique sur les programmes de pré-retraite du CERN et facteurs de la Caisse de pensions

22–23 Jan | Computing | CERN Science Gateway | Voxxed Days | EN
Voxxed days CERN (limited number of free places for CERN personnel)

25 Jan 10:00 | At CERN | Main Auditorium and online | Directorate | EN and FR
New Year presentation to personnel | Présentation du Nouvel An au personnel

25 Jan 16:30 | Knowledge Sharing | CERN Library | Library events | EN
Meet the author - Chris Quigg about 'Grace in All Simplicity'

26 Jan 11:00 | At CERN | Online | Alumni | EN
CERN Alumni: Virtual Company Showroom with Sky Italia

26 Jan 13:00 | At CERN | 774 Auditorium, Prévessin | Internal Communications | EN
Film screening of 1983 BBC Horizon documentary "The Geneva Event"

26 Jan 14:00 | Knowledge Sharing | IT Amphitheatre | HR Learning & Development | FR
L&D Micro-talk 8 - Ego vs innovation : le regard des neurosciences cognitives

30 Jan 10:30 | At CERN | Online | CERN Service Management | EN and FR
5th CERN Service Management Forum

30 Jan 11:00 | Computing | Council Chamber | CERN Computer Security | EN
CERN Computer Security: Abuse, Blunder and Fun

30 Jan 15:00 | Knowledge Sharing | CERN Science Gateway | Arts at CERN | EN
Unveiling the Universe: Art and Science Summit – 70 years of discoveries at CERN

30 Jan 17:30 | Knowledge Sharing | Council Chamber | Staff Association | FR
Projection film : CE QUI NOUS SAUVE

31 Jan 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN
50 years of QCD

1 Feb 12:00 | At CERN | Online | Medical Service | EN and FR
Cancer awareness campaign (more information)

Registration now open

9–11 Feb | Knowledge Sharing | CERN Science Gateway | Alumni Relations
CERN Alumni 3rd collisions

11–14 Feb | Accelerators | Wiener Neustadt (AT)| MedAustron, GSI, I.FAST | EN
5th Slow Extraction Workshop

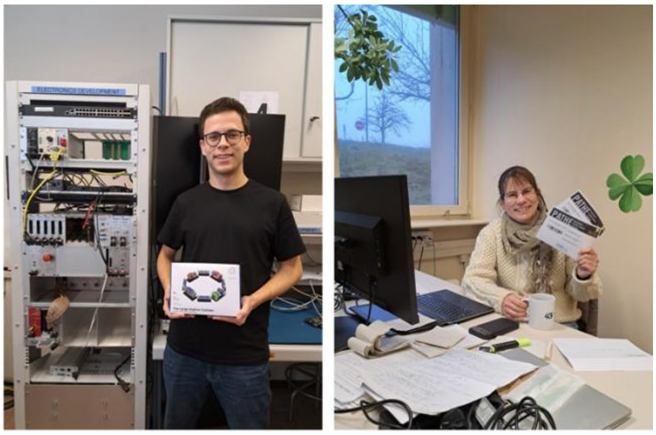
2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN
Mechanical & Materials Engineering for Particle Accelerators and Detectors

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN
2024 Asia-Europe-Pacific School of High-Energy Physics

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN
Accelerators for healthcare? 10-day innovation challenge

This is a curated list of events relevant to the CERN community.
More events are available here: home.cern/events
Indico also shows ALL events happening today, this week and in a calendar view.
If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

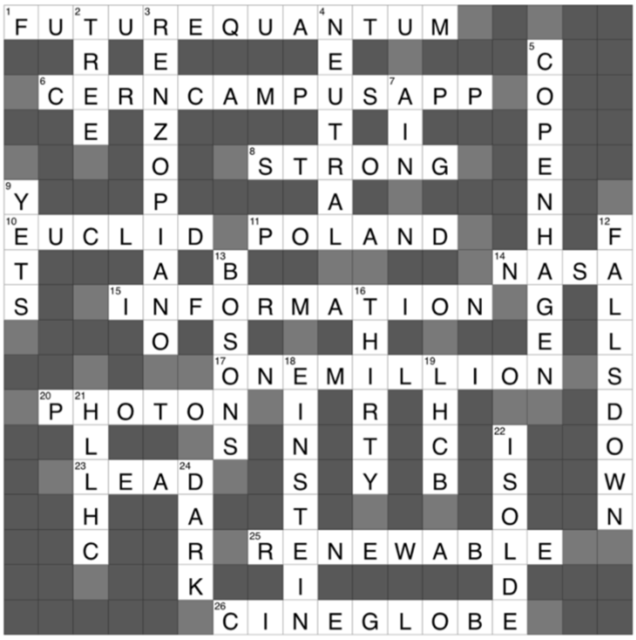
CERN Community: thanks for taking part in the 2023 end-of-year crossword!



Congratulations to the three prize draw winners! André Pinho (left) won the LHC Lego set, Gabriele Thiede (not pictured) won the two [CAGI Geneva Choco Passes](#) and Veronique Lefebure (right) won the two [CAGI cinema tickets](#). (Image : CERN)

Thank you to all who participated in the first [end-of-year crossword](#) for the CERN Community. We hope you had fun and enjoyed reflecting on the past year at the Laboratory with us.
Congratulations to André Pinho (SY-EPC), Gabriele Thiede (FAP-DHO) and Veronique Lefebure (IT-CS), who all had correct entries and were selected in the prize draw.

A special mention to Florian Rehm (BE-CSS), who in a [workshop on Applied AI at CERN](#), used the crossword as an example of how a CERN-developed "AccGPT" could access information.
The full solutions can be found here:



CERN Internal Communication

Work on the Route de Meyrin: traffic disruptions

From 24 January to 26 March, traffic between the Meyrin border crossing and the Porte de France roundabout will be disrupted on and outside the CERN site

Please note that traffic will be disrupted both **on and outside** the CERN site between the Meyrin border crossing and the Porte de France roundabout while work is carried out to replace the perimeter fencing and clear areas of land.
This work will take place in three phases **between 24 January and 26 March 2024**:
Phase 1: 24 January–14 February
The first phase concerns the stretch of perimeter fencing between Buildings 878 and 298 (sector 1, see map). A temporary foot and cycle path will be created on the site, replacing the pavement of the D984F (Route de Meyrin). Temporary fencing will be erected along this path and **Route Schrödinger will be for one-way traffic only** between Route Einstein and Building 298. **Traffic on the Route de Meyrin will not be affected.**
Phase 2: 15 February–8 March
The second phase concerns the stretch of perimeter fencing between Building 878 and the tunnel linking the Meyrin and Prévessin sites (sector 2, see map). A temporary foot and cycle path will be created on the site, replacing the pavement of the D984F (Route de Meyrin).

Temporary fencing will be erected along this path and the stretch of **Route Schrödinger between Route Einstein and Building 878 will be closed to traffic. Traffic on the Route de Meyrin will not be affected.**

Phase 3: 11–26 March

The third and final phase concerns the stretch of perimeter fencing between the border crossing and the tunnel linking the Meyrin and Prévessin sites (sector 3, see map). This work will be carried out at night, between 8.00 p.m. and 6.00 a.m., when a contraflow system controlled by traffic lights will be in operation. A foot and cycle path will remain in place throughout the work. Traffic will not be affected during the day.

Details of the work, including any changes to the work schedule, will be displayed on the information panels on the Meyrin site.

Thank you for your understanding.

SCE department

Living and working with winter viruses circulating: taking care of yourself and others

In these colder winter months, we naturally spend more time indoors and in close proximity to one another. These are perfect conditions for the rapid spread of seasonal respiratory illnesses, such as the common cold, flu and COVID. Preventive measures and self-care are key at this time to ensure we stay in good health and avoid contaminating others.

In the event of a mild seasonal illness (COVID, mild flu, cold, etc.), with symptoms such as a cough or runny nose in particular, it is important to respect a few basic rules in order to preserve your own health, but also that of others. The rule of thumb is to follow the seven tips to prevent the spread of viruses shown in the poster displayed around the site and reproduced below. Out of respect for others, contact with colleagues should be avoided as much as possible and, if this is not possible, wearing a mask, keeping your distance and frequently ventilating the workspace is key. Contaminating colleagues does no one any favours, including the Organization.

CERN provides the following options in this context:

- Three consecutive days of sick leave without a medical certificate are tolerated, up to a limit of seven days per year.
- To the extent that it is compatible with their work, Operational Circular (OC) 7 allows members of the personnel to work from home for 40% of their working time over a period of two weeks. This may be extended with the agreement of the supervisor in specific circumstances (including being indisposed for a prolonged period when suffering from a viral infection).

Finally, nowadays, meetings can easily be held in virtual or hybrid mode.

Take good care of yourself: get plenty of rest and, if symptoms persist, it is best to consult a doctor.

HSE unit

CERN Medical Service: Making healthy choices to cut cancer risk

Join the cancer awareness and prevention campaign run by the CERN Medical Service and several partners on 1 February

4 February marks World Cancer Day, an initiative started in 2000 and led by the Union for International Cancer Control (UICC).

Cancer is a global health challenge that affects millions of people. It is a complex group of diseases characterised by the uncontrolled growth and spread of abnormal cells in the body. The World Cancer Research Fund (WCRF) estimates that as many as 18.1 million cases were diagnosed worldwide in 2020, with near equal impact on men and women. The most common types of cancer include breast, lung, gastrointestinal and prostate cancers – though there are many more – and cancer is estimated to have caused roughly one in six deaths in 2020 (WHO). While the prevalence of cancer can vary by region, type of cancer and demographic factors, cancer is unequivocally a leading cause of morbidity and mortality.

There is hope, however: the WHO suggests that up to 50% of cancers can be prevented, either by making lifestyle changes or through simple proactive medical interventions. By making informed choices, starting with adopting an active lifestyle and a varied, healthy diet, avoiding tobacco and alcohol consumption, minimising exposure to the sun and undergoing screenings and vaccinations where applicable (e.g. HPV and hepatitis B), individuals can reduce their cancer risk and contribute to the global effort to combat the disease.

To raise awareness and inform the CERN community of what we can each do to reduce our own risk of developing cancer, and how to support those who suffer from the disease, the CERN Medical Service is organising a dedicated campaign on **Thursday, 1 February**.

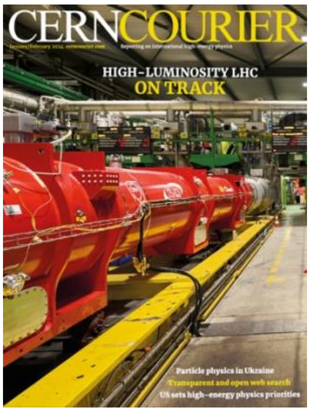
In **Restaurant 1, from 10 a.m. until 3 p.m.**, the CERN Medical Service, along with representatives from *La ligue genevoise contre le cancer*, *La ligue française contre le cancer*, Cancer Support Switzerland and la *Fondation genevoise pour le dépistage du cancer*, as well as the CERN psychologists, will be running a cancer awareness stand. The Knowledge Transfer group will also be present to demonstrate advancements in cancer therapy that use CERN innovation and technology.

At **12 p.m., a Zoom webinar** will feature microtalks by the Hôpitaux Universitaires de Genève (HUG) and the abovementioned partners. These will span epidemiology, risk factors, treatment and prevention measures and advice on how to manage the disease, both for cancer patients and for those supporting them. The CERN psychologists will also give advice about living and working with cancer and the impact of stress in this context.

Details of the webinar can be found here. Further information and resources can be found on the Medical Service's dedicated cancer awareness webpage.

Medical Service

The January/February issue of the CERN Courier is out



With just under two years of LHC operations remaining before the collider is shut down to make way for its high luminosity upgrade (HL-LHC), 2024 is a big year for teams across CERN and beyond. The focus now is on the validation of key technologies, tests of prototypes and the series production of equipment (p37).
Due to deliver high-brightness beams from 2029, the HL-LHC will bring rich physics opportunities for the four main experiments into the early 2040s.
The experience gained from the HL-LHC will also be key to the success of a future collider at CERN. On that note, a February session of the CERN Council is to assess impressive progress documented in the mid-term review of the Future Circular Collider feasibility study. Meanwhile, in December the US "P5" report expressed strong support for a Higgs factory in Europe or Japan (p7). December also brought news from the CERN Council that CERN's cooperation with Russia and Belarus will conclude at the expiry of their respective international collaboration agreements: 30 November and 27 June 2024. This issue looks at how the war has impacted particle physics in Ukraine through the experiences of researchers at institutes in Kharkiv, Kyiv, Odesa and Uzhhorod (p30). The lure of the CERN model (p43), 3D-printed detectors (p9), fair and transparent web search (p45), 40 years of the CERN Accelerator School (p25), and how to lead in collaborations (p50) are other must-reads.
Read the digital edition of this new issue on CDS.

CERN agreement with Elsevier in 2024

The CERN Library is a member of the Swiss Consortium of Academic Libraries, which works closely with SwissUniversities to negotiate agreements with scholarly publishers on behalf of universities and research institutions in Switzerland, and includes CERN in some of these negotiations.
Recently, the negotiating team of SwissUniversities informed CERN that discussions with Elsevier are not making progress and there is a high risk that no agreement will be reached by January 2024, when our current agreement will expire.
The agreement with Elsevier includes two components: the subscription component to access the content of their journals ("Read"), and the Open Access publishing of articles from participating institutions' authors in the journals ("Publish")
As a consequence, CERN users may lose access to new content published by Elsevier after January 1st, 2024. All content published before 2024, as well as all content published Open Access remains available, e.g. journals funded by SCOAP3: Nuclear Physics B and Physics Letters B. The CERN Library is committed to provide the content by other means, so if you need an article that is not available, please fill this form. We will do our best to send you the article within 24 hours. In addition, SwissUniversities has put together a document that describes other ways of accessing content included in those journals. Please have a look at this fact sheet.
The systematic Open Access publishing of CERN articles in Elsevier journals (outside of Nuclear Physics B and Physics Letters B) enabled through our current agreement will cease on January 1st, 2024. This means that articles accepted in Elsevier journals after this date will not automatically be published Open Access supported by the central fund of the CERN library.
However, CERN authors and collaborations are required to comply with the CERN Open Access policy. Please note that CERN has valid Open Access agreements with many other publishers, you can find the list here. If you have an article

already submitted or about to be submitted in one of the Elsevier journals, please reach out to us for advice.

We will inform you as soon as the situation changes. The negotiations with Elsevier will continue in 2024, to hopefully achieve a mutually acceptable agreement.

In the meantime, do not hesitate to contact us at: open-access-questions@cern.ch for any questions or concerns.

CERN Library

Obituaries

Bruce Marsh (1980 – 2023)

It is with deep sadness that we share with you the tragic loss of our much-esteemed friend and colleague, Bruce Marsh, Leader of the Lasers and Photocathodes section in the Sources, Targets and Interactions group within the Accelerator Systems department. He was an invaluable member of our group for many years, contributing to our shared endeavours with his vast expertise, knowledge, attention to detail and kindness towards the people around him.

Bruce completed his PhD at the University of Manchester, and his thesis research included work at CERN, where he later returned as a fellow in 2006 before taking up a staff position in 2010. In his different roles, Bruce contributed greatly to the development of the resonance ionisation laser ion source techniques dedicated to the production of radioisotopes for fundamental research and medicine at the ISOLDE facility. His involvement extended beyond CERN's boundaries, influencing the global landscape of nuclear and laser physics research. Bruce's work consistently achieved international recognition, with some results being published in the very prestigious journal Nature. He was invited to present his work at numerous international conferences, workshops and schools and was instrumental in the organisation of many such events.

Furthermore, Bruce worked closely with his team to contribute to the advancement of future accelerators and colliders at CERN, with a particular focus on CLIC, AWAKE and the Gamma Factory. His dedicated efforts thus contributed to the long-term future of the Organization.

Bruce also achieved international recognition through his role as general coordinator of the EU-funded Marie Curie training network Laser Ionisation and Spectroscopy of Actinides (LISA), comprising some 12 laboratories in Europe and several other international partners.

Bruce will be fondly remembered for his warm and welcoming spirit, always ready with a smile for everyone around him, and for his strong sense of justice. An unwavering champion of diversity in

the workplace, Bruce fostered an inclusive environment where every voice was valued and every individual felt empowered to contribute with their unique perspectives. He provided invaluable opportunities for integration, learning and professional growth for his colleagues.
During these difficult times, our thoughts and sincerest condolences go out to Bruce's family, his friends, and all those who, like us, were lucky enough to have known him.
He will be greatly missed and cherished by all of us who had the privilege to work alongside him.

His friends and colleagues

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We are deeply saddened to announce the death of Mr Bruce Alan Marsh on 30 December 2023.
Bruce Alan Marsh, who was born on 10 June 1980, worked in the SY department and had been at CERN since 1 February 2003.
The Director-General has sent a message of condolence to his family on behalf of the CERN personnel.

Social Affairs service
Human Resources department

Ombud's corner

Active listening – from sponge to trampoline

Linda* makes a deliberate effort to be an active listener in a discussion with Igor*. She gives time to the discussion with him and puts all distractions to one side. She listens carefully to what Igor says and nods consciously to demonstrate understanding and attention. She even remembers to reformulate in order to show that she has understood his message. She bounces off what Igor says and gives examples based on her own experience.
Despite all Linda's efforts to be an active listener, Igor leaves the discussion feeling unheard and close to feeling dismissed. What went wrong?
In this article, I would like to share with you how difficult active listening can be and how we can improve this skill.
Back to basics: Active listening is when you not only hear what someone is saying, but are also attuned to their thoughts and feelings. If you actively listen to someone, you are turning a conversation into an active, non-competitive, two-way interaction.
Each of us has a natural listening style:

- A *task-oriented listener* is focused on the efficient transfer of important information.
- An *analytical listener* is primarily analysing a problem.
- A *relational listener* tries to build a connection and respond to the speaker's emotions.
- A *critical listener* judges the content of the conversation and, possibly, the speaker themselves.

I admit to being, in my everyday interactions, a mixture of an analytical and critical listener, mostly inherited from previous functions, but, in my role as Ombud, I make a conscious effort to be an active listener when visitors share with me the challenging situations they are facing.
The art of active listening requires a conscious and deliberate effort on three levels:
Cognitive: you have to devote all your attention to what the other person is saying, either explicitly or implicitly, and you have to understand and integrate this information.
Emotional: you must manage your own emotions during the conversation. Many emotions may well

up inside you during a conversation: enthusiasm, surprise, excitement, but also annoyance, boredom or even anger. You must be conscious of your own emotions, as they have no place in active listening.

Behavioural: you need to convey interest and understanding, verbally and non-verbally.

As Ombud, I very much like and fully adhere to this metaphor[1] of an active listener acting as a trampoline:

"You're not a sponge merely absorbing information. Instead, think of yourself more like a trampoline that gives the speaker's thoughts energy, acceleration, height, and amplification."

Still, it's far from easy to go from simply absorbing information – like a sponge – to helping your colleague abandon an inefficient thought process and bounce back. One way to achieve this and ensure that you actively listen to the ongoing conversation is to ask yourself simple questions:

- Why do I need to listen right now?

Reflecting on the purpose of the conversation, both for you and for your colleague, will help you to decide which type of listener you need to be at that moment. Empathy and consideration of the other's needs will help you tune your listening style.

- Who is the focus of attention?

This is another pitfall to be aware of, especially for the Ombud. We are all tempted to quote stories from our personal experience, but this steers the conversation away from the needs and perceptions of your interlocutor. It takes experience to resist this temptation, which is detrimental to active listening.

- Am I really listening?

Very often, in our everyday conversations, we think about what we're going to say next to our colleague, or what question we're going to ask. This is no longer listening. We may also be busy thinking of ways to convey attention: Should I nod now? Unfold my arms? Smile? Lean forward? And so on.). In both cases, we are no longer listening.

- Am I still listening?

Especially in those cases where our interlocutors are rather talkative, we tend to think that we have heard enough to form an opinion, stop listening and surreptitiously check our smartphones. We are no longer listening, and they may be saying something that is really key to helping them unlock the situation.

- What could I be missing that my colleague is not telling me?

Active listening is all about asking the right questions, which will help your interlocutors to reflect on their issue and how they are approaching it. They may also give important non-verbal clues, such as how uncertain, vulnerable, confident or confused they may feel. These clues will help you to adapt your questions.

Reflecting on your natural listening style and asking yourself these simple questions will help you to practise effective active listening. It takes extensive experience to become an active listener, but your interlocutor will leave the conversation feeling heard and understood, which is the first step to overcoming a challenging issue. Moreover, research shows that those who engage in active listening are seen as more competent, likeable and trustworthy by others.

Allow me to wish you all a very happy, healthy and peaceful 2024, under the auspices of open-mindedness, empathy and true collaboration!

Laure Esteveny

** Names are fictitious*

This article is inspired by an article by Amy Gallo published in the Harvard Business Review journal, 2 January 2024: "What is active listening?"

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at https://mattermost.web.cern.ch/cern-ombud/.

More information on the role of the CERN Ombud and how to contact her can be found at https://ombud.web.cern.ch.

[1] *"What great listeners actually do", Jack Zenger etJoseph Folkman, Harvard Business Review, 14 juillet 2016.*